

WITS
UNIVERSITY

Not All Sparsity Is Equal: A Controlled Evaluation of Recommender Systems under Limited User Feedback

Kamal Bhupendra Lalloo, Student No. 2652159
School of Computer Science and Applied Mathematics, Faculty of Science, University of the Witwatersrand
Supervisor: Dr. Hairong Bau 2026

***Abstract*—Recommender systems rely on historical user interactions to infer preferences, but in practical settings this feedback is often limited. This study investigates how recommendation performance changes when user histories become sparse, with particular attention to whether the temporal position of retained interactions affects robustness. Ten recommender systems were evaluated on MovieLens-1M and Amazon Video Games using three controlled sparse-feedback mechanisms: random per-user, recent-history, and early-profile retention at nominal 50%, 25%, and 10% of each user's training history. Validation and test interactions were held fixed across conditions, and models were evaluated using full-sort ranking with Normalized Discounted Cumulative Gain at 10 as the primary metric.**

Performance generally declined as feedback was reduced, but the extent of degradation depended strongly on the retention mechanism and dataset. Recent-history retention preserved substantially more recommendation quality than random or early-profile retention, particularly on MovieLens, while Amazon remained more sensitive under severe sparsity, alongside substantially shorter retained user histories and greater loss of item exposure. SASRec and GRU4Rec achieved the strongest absolute performance across most conditions, although greater architectural complexity did not consistently imply greater robustness. Overall, the findings show that sparse-feedback robustness depends not only on how much interaction data remains, but also on which interactions remain and the characteristics of the dataset.

***Index Terms*—recommender systems, data sparsity, implicit feedback, sequential recommendation, robustness evaluation, temporal retention**

I. Introduction

Recommender systems are widely used to help users navigate large collections of items by learning from their previous interactions. In many practical settings, recommender systems learn from implicit behavioural signals such as clicks, purchases, or views, while explicit ratings can also be transformed into positive interaction signals for implicit-feedback modelling [1]. Modern recommender systems use a wide range of approaches to exploit this information, from neighbourhood and latent-factor methods to neural, sequential, and graph-based models. Despite their different architectures, these methods share an important dependency: useful recommendations generally require sufficient historical information about users and items.

In practice, however, interaction histories are often sparse. Some users interact with only a small number of items, while even active users may provide only limited evidence about their current preferences. This creates a challenge for collaborative recommendation because models must infer preferences from increasingly incomplete information. Previous work has shown that recommendation performance can deteriorate as interaction data becomes sparse [2]. At the same time, methodological studies have demonstrated that conclusions about recommender performance can depend strongly on choices such as preprocessing, data splitting, baseline tuning, and evaluation procedure [3], [4]. Consequently, a model that performs strongly on a conventional full-data benchmark cannot automatically be assumed to remain strong when the available feedback is substantially reduced.

A further issue is that sparsity is not characterised only by the *number* of available interactions. User histories are temporally ordered, and interactions at different positions may contain different amounts of information about a user's current preferences. Betello et al. [5], for example, showed that sequential recommender performance is sensitive to the position of interactions removed from training sequences. This raises a broader question: when a user history becomes sparse, does it matter which portion of that history remains? Comparing models only under full interaction histories, or reducing data without considering temporal position, may therefore hide important differences in how recommender systems respond to limited feedback.

This study investigates this problem through a *controlled sparse-feedback robustness evaluation*. Ten recommender systems spanning non-personalised, classical, linear, neural, sequential, and graph-based approaches are evaluated on MovieLens-1M and the Amazon Reviews 2023 Video Games dataset. For each dataset, a fixed temporal validation and test split is created before the training histories are reduced. Three sparsification mechanisms are then applied at nominal retention levels of 50%, 25%, and 10%: *random per-user retention*,

which retains interactions from across the training history; *recent-history retention*, which preserves the most recent interactions; and *early-profile retention*, which preserves the earliest interactions. The same held-out targets and candidate catalogue are maintained across conditions so that differences can be attributed more directly to the information retained in the training histories. This controlled design also follows recent calls for stronger methodological standards and transparent offline recommender evaluation [6].

Accordingly, the study examines how recommendation performance changes as the available user feedback decreases; whether performance under sparse user histories depends on the temporal location of the retained feedback; and whether more complex recommender approaches are consistently more robust than simpler alternatives.

The work makes three main contributions: (1) a controlled evaluation protocol that varies both the quantity and temporal location of retained user feedback while holding evaluation targets and the item catalogue fixed; (2) a cross-paradigm comparison of ten recommender systems under identical sparse-feedback conditions; and (3) empirical evidence showing that sparse-feedback behaviour depends jointly on temporal position, model choice, and dataset characteristics, rather than on interaction quantity alone.

Results show that recent-history retention generally preserves more recommendation quality than equally sized random or early-profile subsets, although the advantage is strongly dataset-dependent. SASRec and GRU4Rec achieve the strongest absolute performance across most conditions, while greater complexity does not consistently imply greater robustness.

The remainder of this paper is organised as follows. Section II reviews related work; Section III describes the data and methodology; Section IV presents the results and statistical analysis; Section V discusses the findings, limitations, and future work; and Section VI concludes.

II. Related Work

A. Recommender Approaches and Sequential Feedback

Recommender systems commonly learn user preferences from explicit or implicit interaction data. In implicit-feedback settings, user actions such as purchases, clicks, or positive ratings provide evidence of preference without directly observing negative feedback [1]. Classical collaborative-filtering approaches model these interactions through neighbourhood relationships or latent representations, with methods such as Bayesian Personalized Ranking (BPR) providing a widely used framework for learning from implicit preferences [7]. These approaches generally treat a user's interaction history as a set of preferences rather than explicitly modelling when each interaction occurred.

Sequence-aware recommendation extends this formulation by treating interaction order as informative [8]. Recurrent approaches such as GRU4Rec [9] model evolving behaviour through recurrent hidden states, while SASRec [10] uses self-attention to identify relevant interactions within a user's history. BERT4Rec [11] further applies bidirectional Transformer-based sequence modelling. Graph-based methods provide another approach to collaborative information, with LightGCN [12] propagating user and item representations through the interaction graph. Together with neural collaborative-filtering and autoencoder approaches, these developments provide increasingly diverse ways of using interaction history. However, stronger model complexity does not necessarily imply consistently better recommendation performance, particularly when comparisons involve differently tuned baselines or experimental protocols.

B. Rigorous and Reproducible Evaluation

The reliability of offline recommender comparisons has become an important research concern. Neural methods do not consistently outperform well-tuned simpler baselines [3], while preprocessing, tuning, and other evaluation choices can materially affect comparisons [4]. DaisyRec 2.0 formalised these evaluation factors [13], and recent guidance continues to emphasise tuned baselines, transparent procedures, and reproducibility [6].

Evaluation choices also affect the meaning of reported ranking metrics. Krichene and Rendle [14] showed that metrics calculated using sampled candidate sets can be inconsistent with their exact full-catalogue counterparts and recommended avoiding sampling where feasible. For sequential recommendation, the way interactions are divided between training and evaluation is similarly important. Gusak et al. [15] demonstrated that different temporal splitting strategies can change model rankings and noted that commonly used leave-one-out protocols may not fully represent a globally future-facing deployment scenario. These findings motivate clearly specifying both the candidate-ranking procedure and the temporal evaluation protocol when comparing recommender robustness.

C. Sparse Feedback and Temporal Robustness

Data sparsity has long been recognised as a challenge for collaborative recommendation because each user typically interacts with only a small fraction of the available catalogue. da Silva et al. [2] directly evaluated collaborative-filtering algorithms under different sparsity levels and showed that changes in available interaction data can materially affect recommendation performance. Much sparse-recommendation research has therefore focused on developing models that can learn effectively from limited observations. However, interaction count alone does not capture the temporal structure present in user histories.

Controlled feedback reduction is distinct from conventional cold start, which concerns users or items with little or no history. Here, known users are retained while portions of their existing histories are systematically reduced; early-profile retention is therefore cold-start-like rather than true new-user cold start.

Recent work has begun to examine robustness to missing or perturbed sequential interactions more directly. Betello et al. [5] found that removing interactions near the end of ordered sequences had a substantially larger effect than removing earlier interactions, showing that temporal location can be as important as the amount removed. Where their work studies perturbations within sequential recommenders, the present study applies temporally controlled feedback reduction across ten models

spanning non-personalised, classical, linear, neural, sequential, and graph-based approaches.

Standardised reproduction has further shown that sequential-model conclusions, including SASRec versus GRU4Rec comparisons, can depend on experimental configuration and model capacity [16].

D. Summary and Positioning

Existing research therefore establishes three relevant points: recommendation performance is sensitive to data sparsity; temporal information can affect which interactions are most useful; and experimental design can materially influence model comparisons. However, these issues are often examined separately or within a restricted model family. The present study builds on this work through a controlled cross-paradigm evaluation of ten recommender systems, varying both the quantity and temporal location of retained feedback across MovieLens-1M and Amazon Video Games. This positions the study as an evaluation of recommender robustness under controlled feedback reduction rather than as the proposal of a new recommendation algorithm.

III. Methodology

A. Experimental Design

This study uses a controlled experimental pipeline to evaluate how recommender systems respond when the amount and temporal position of available user feedback are changed. The central design principle was to vary the training history available to each model while keeping the recommendation task itself fixed. After preprocessing, each user's interactions were ordered chronologically and divided using temporal leave-one-out: the final interaction was used for testing, the second-last for validation, and all preceding interactions for training. Sparsification was then applied only to the training portion. Consequently, the validation and test targets remained identical across the full-data baseline and all sparse conditions. The post-preprocessing item catalogue was also kept fixed so that differences between conditions were not caused by changing candidate sets.

For sequential recommenders, the sequence preceding the fixed test target was constructed from the retained sparse training history together with the chronologically available validation interaction; removed training interactions were not restored. Non-sequential recommenders likewise learned their user/item representations, interaction matrices, or graphs only from the retained sparse training interactions.

The validation interaction precedes the test target and is therefore included in the sequential models' test-time sequence; this reflects sequence-based inference, while sparsified training histories and held-out targets remain fixed across conditions.

Figure 1 summarises the experimental pipeline. Hyperparameters were tuned separately for MovieLens and Amazon on the full-data baseline, using validation NDCG@10 to select each model's configuration. The selected dataset-specific configuration was then frozen and reused at 50%, 25%, and 10% retention under all three mechanisms. The experiment therefore measures the robustness of a common baseline-selected configuration, rather than the best performance obtainable by retuning for each sparse condition. Appendix A reports the selected configurations.

B. Datasets and Preprocessing

Experiments were conducted on MovieLens-1M [17] and the Amazon Reviews 2023 Video Games subset [18]. MovieLens and Amazon were selected for their contrasting interaction-history lengths, allowing sparse-feedback behaviour to be compared across substantially different history regimes.

Explicit ratings were converted to positive implicit feedback by retaining interactions with ratings of at least four. For Amazon, repeated positive user-item pairs were deduplicated by retaining the most recent positive interaction; 50,578 repeated rows were removed. No retained positive duplicates were present in MovieLens. Both datasets were subjected to iterative 5-core filtering, ensuring at least five interactions per remaining user and item before sparsification.

Interactions were ordered first by timestamp and then by their original source-row order to deterministically resolve timestamp ties. A sequential order field was assigned after sorting and used throughout temporal splitting and sequential-model training. Table I reports the final dataset and split statistics.

The mean post-filtering history length was 95.19 interactions for MovieLens and 8.41 for Amazon. After holding out validation and test interactions, users therefore had an average of 93.19 training interactions on MovieLens but only 6.41 on Amazon. This difference is important because an equivalent nominal reduction in user history does not leave the same amount of usable evidence in the two datasets.

C. Sparse-Feedback Construction

Three sparsification mechanisms were applied independently at nominal retention levels of 50%, 25%, and 10%, spanning moderate to severe feedback reduction alongside the 100% baseline.

Random per-user retention samples interactions from across the user's training history and is intended to reduce interaction quantity without deliberately favouring a particular temporal region. A fixed sparsity seed of 2025 was used, and the retained subsets were nested so that the 10% subset was contained within the 25% subset, which was contained within the 50% subset.

Recent-history retention preserves the most recent training interactions and tests whether a shorter but temporally relevant history can retain sufficient information for predicting the user's next interaction.

Early-profile retention preserves the earliest training interactions. This represents a cold-start-like setting for known users, but it is not a true new-user cold-start experiment because the same users are present in training. It also increases the temporal distance between the retained training interactions and the held-out test target.

For a user u with n_u training interactions and nominal retention rate p, the number retained was

$$m_u(p) = \max\left(1, \lceil n_u p \rceil\right). \tag{1}$$

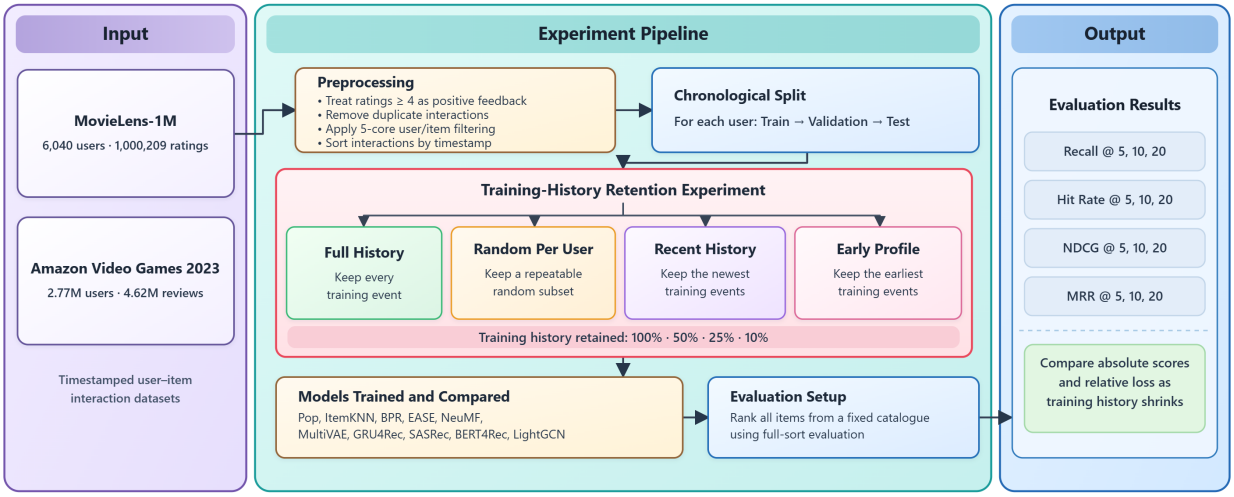


Fig. 1. Experimental pipeline. Explicit feedback is converted to positive implicit interactions, filtered and chronologically ordered before a per-user temporal leave-one-out split. Only the training histories are sparsified using random per-user, recent-history, or early-profile retention. Models are tuned on the full-data baseline, the selected configurations are frozen, and all conditions are evaluated using the same validation/test targets and fixed item catalogue.

TABLE I
FINAL DATASET AND TEMPORAL SPLIT STATISTICS.

Dataset	Users	Items	Interactions	Train	Mean train/user	Median history
MovieLens-1M	6,034	3,125	574,376	562,308	93.19	58
Amazon Video Games	63,413	19,022	533,550	406,724	6.41	6

The minimum of one retained interaction in (1) prevents users from disappearing from the training set. As a result, nominal retention differs from aggregate realised retention, particularly for Amazon because its histories are much shorter. MovieLens realised retention was 50.270%, 25.404%, and 10.479%, while Amazon realised retention was 54.712%, 29.767%, and 18.380%. At the nominal 10% setting, this corresponds to an average of 9.76 retained training interactions per MovieLens user but only 1.18 per Amazon user.

D. Recommender Models and Training

Ten recommenders were selected to cover a range of commonly used recommendation paradigms rather than focusing on a single model family. These were the non-personalised popularity baseline Pop; ItemKNN [19] and BPR [7] as classical approaches; the linear EASE model [20]; NeuMF [21] and MultiVAE [22] as neural collaborative-filtering approaches; GRU4Rec [9], SASRec [10], and BERT4Rec [11] as sequential recommenders; and LightGCN [12] as a graph-based recommender.

Experiments were implemented using RecBole 1.2.1 [23]. A local BERT4Rec compatibility patch corrects masked-target validity handling so that a legitimate masked target at sequence position zero is not discarded; it does not change the architecture, forward pass, masking procedure, prediction logic, or evaluation behaviour. The maximum sequence length was 200 interactions for MovieLens and 50 for Amazon.

Stochastic models (BPR, NeuMF, MultiVAE, GRU4Rec, SASRec, BERT4Rec, and LightGCN) were evaluated using seeds 2025, 2026, and 2027, while deterministic models (Pop, ItemKNN, and EASE) were run once per condition. Code, experimental configurations, preprocessing scripts, tests, and analysis materials are available in the project repository.[1]

E. Evaluation Metrics

All models were evaluated using RecBole full-sort ranking rather than sampled-negative evaluation [14]. Each user's held-out test interaction was ranked against the fixed candidate catalogue. Results were recorded for Recall, HitRate, NDCG, and MRR at $K \in \{5, 10, 20\}$, with NDCG@10 used as the primary metric because it captures both successful retrieval and the rank assigned to the relevant item.

Because temporal leave-one-out produces exactly one relevant test item per user, the user-level NDCG contribution at cutoff K [24] can be written as

$$\mathrm{NDCG@}K = \begin{cases} \dfrac{1}{\log_2(r+1)}, & r \le K, \\ 0, & r > K, \end{cases} \tag{2}$$

where r is the rank of the held-out item. Under the same design, Recall@K and HitRate@K are numerically identical and are therefore not interpreted as independent evidence.

To compare robustness between models with different baseline accuracies, relative NDCG change for model m in sparse condition c was calculated as

$$\Delta_{m,c} = \frac{S_{m,c} - S_{m,\mathrm{full}}}{S_{m,\mathrm{full}}} \times 100. \tag{3}$$

[1] https://github.com/KamalLalloo/recommender-sparsity-honours

Here, $S_{m,c}$ is sparse-condition NDCG@10 and $S_{m,\text{full}}$ is the corresponding full-data score. Negative values from (3) indicate degradation and positive values improvement. Relative changes are considered together with absolute NDCG because a model can appear relatively stable while still providing low recommendation quality.

F. Statistical Analysis

Statistical analysis used aggregate model-condition results, averaging the three seeds of each stochastic model so that training repeats were not treated as independent observations; deterministic models contributed one result. The units entering the tests were therefore aggregate scores for the evaluated models and conditions, rather than per-user held-out ranks or individual training runs. Four comparison dimensions were defined. *Overall model comparison* asked whether the ten models differed within a dataset, treating the ten experimental conditions as related blocks; it used a Friedman omnibus test, Kendall's W, and, when significant, Nemenyi comparisons of average ranks [25]. *Mechanism comparison* held dataset and retention fixed and paired Random, Recent, and Early scores across the same ten models. *Retention comparison* held dataset and mechanism fixed and paired 50%, 25%, and 10% scores across those models. Both used Friedman tests followed, where needed, by focused Wilcoxon signed-rank tests with Holm correction [26]. *Cross-dataset agreement* used Spearman correlation to compare full-data model ranks and relative-robustness ranks. Friedman and Wilcoxon tests provide non-parametric comparisons across the matched model-condition observations used here without requiring normally distributed scores. Because the experimental conditions share users, targets, and related training histories, the resulting inference is interpreted at the aggregate model-condition level rather than as independent user-level evidence. All tests used $\alpha = 0.05$.

The tests therefore assess systematic differences across the evaluated models and experimental conditions without extending the claims to recommender algorithms beyond those considered here.

IV. Experiments and Results

A. Experimental Setup

Ten recommender systems were evaluated on MovieLens-1M and Amazon Video Games under the full-data baseline and nine sparse-feedback conditions. Random per-user, recent-history, and early-profile retention were each evaluated at nominal retention levels of 50%, 25%, and 10%. NDCG@10 is the primary metric, with stochastic-model results reported as means across seeds 2025, 2026, and 2027. The complete experimental protocol is defined in Section III.

B. Full-Data Baseline Performance

Table II presents baseline NDCG@10 performance before any training interactions were removed. SASRec achieved the highest baseline score on both datasets, followed by GRU4Rec. On MovieLens, the three sequential models formed a clear leading group, with SASRec obtaining 0.1191, GRU4Rec 0.1126, and BERT4Rec 0.1067. The strongest non-sequential model, EASE, achieved 0.0452.

TABLE II
Full-data baseline NDCG@10 performance. Values are mean ± standard deviation for stochastic models and single-run scores for deterministic models.

Model	MovieLens-1M	Amazon Video Games
SASRec	0.1191 ± 0.0023	0.0463 ± 0.0004
GRU4Rec	0.1126 ± 0.0015	0.0449 ± 0.0010
BERT4Rec	0.1067 ± 0.0026	0.0391 ± 0.0003
EASE	0.0452	0.0406
MultiVAE	0.0392 ± 0.0006	0.0346 ± 0.0004
LightGCN	0.0381 ± 0.0007	0.0359 ± 0.0004
BPR	0.0380 ± 0.0012	0.0355 ± 0.0005
ItemKNN	0.0373	0.0241
NeuMF	0.0320 ± 0.0009	0.0243 ± 0.0003
Pop	0.0192	0.0163

The differences were smaller on Amazon. SASRec again ranked first with an NDCG@10 of 0.0463, followed by GRU4Rec at 0.0449 and EASE at 0.0406. Across all 20 dataset-condition combinations considered in the study, SASRec achieved the highest mean NDCG@10 in 17 conditions, while GRU4Rec ranked first in the remaining three.

C. Performance Under Sparse Feedback

Figure 2 compares NDCG@10 as the retained portion of each user's training history decreases. The three sparsity mechanisms produce visibly different performance trajectories on both datasets.

Random per-user and early-profile retention produced substantial degradation as less feedback remained. The mechanism mattered especially on MovieLens at nominal 10% retention: median relative NDCG@10 change across models was only -6.8% for recent-history retention, compared with -74.9% for random per-user and -74.7% for early-profile retention. Amazon was less tolerant: its Recent 10% median change was -63.5%, close to Random 10% (-63.6%) and somewhat better than Early 10% (-68.5%).

D. Behaviour Under Severe Sparsity

Table III reports model-wise performance at nominal 10% retention. Recent-history retention was generally the least damaging mechanism, but it did not make severe sparsity harmless. On MovieLens Recent 10%, SASRec and GRU4Rec lost 54.0% and 54.3% relative to their own baselines; on Amazon, their losses were 40.0% and 52.7%. Thus, the aggregate recent-history advantage coexists with substantial losses for the strongest sequential models. GRU4Rec led Amazon Random 10%, while SASRec led the other five severe dataset–mechanism comparisons.

Relative robustness is not the same as recommendation quality: Amazon Pop barely changed under Recent 10%, yet its absolute NDCG@10 (0.0163) remained well below SASRec (0.0278). Supporting Recall@10 and MRR@10 results show similarly strong rank agreement with NDCG@10 and are reported in Appendix B.

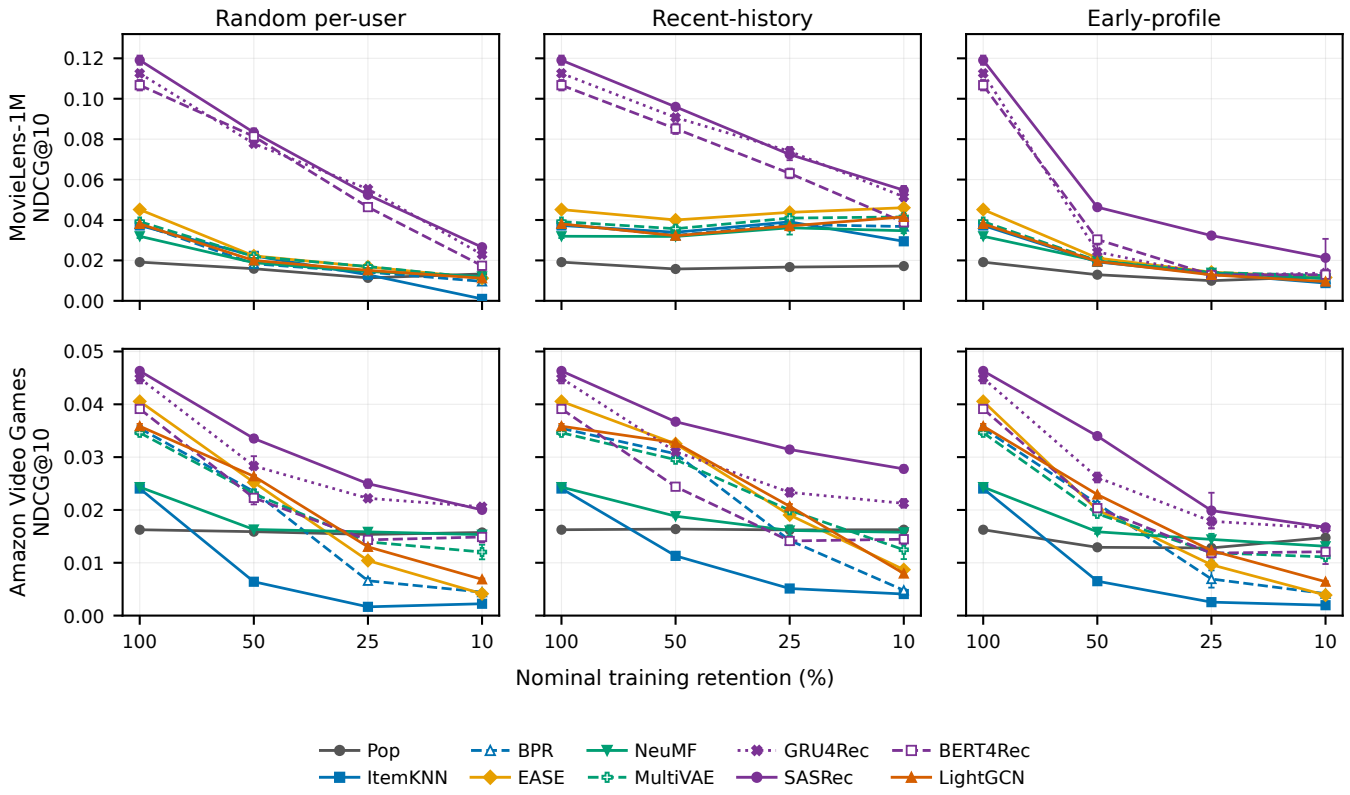


Fig. 2. NDCG@10 trajectories under sparse feedback. Rows correspond to datasets (MovieLens-1M above; Amazon Video Games below), and columns to sparsity mechanisms. The x-axis is nominal training retention and the y-axis is NDCG@10; colours group model families, while line styles and markers distinguish models. Stochastic models show means across three seeds, with error bars denoting one standard deviation.

E. Statistical Evidence

Friedman tests showed significant overall differences in NDCG@10 between the ten models on both datasets. MovieLens produced $\chi^2(9) = 63.03$, $p < 0.001$, with Kendall's $W = 0.700$, while Amazon produced $\chi^2(9) = 59.22$, $p < 0.001$, with $W = 0.658$.

Nemenyi post-hoc comparisons were then used to examine differences in average model rank. On MovieLens, SASRec had significantly better average ranks than LightGCN, NeuMF, BPR, ItemKNN, and Pop ($p < 0.05$), but its rank did not differ significantly from GRU4Rec, BERT4Rec, EASE, or MultiVAE. On Amazon, SASRec significantly outranked BERT4Rec, EASE, MultiVAE, NeuMF, BPR, ItemKNN, and Pop, while its differences from GRU4Rec and LightGCN were not significant.

The sparsity mechanism also had a significant effect at every evaluated retention level on both datasets. The six Friedman tests comparing random, recent, and early retention all produced $p < 0.001$, with Kendall's W ranging from 0.73 to 0.91. Post-hoc Wilcoxon tests showed that recent-history retention significantly outperformed both random per-user and early-profile retention at every retention level on both datasets (Holm-adjusted $p \leq 0.0078$).

When retention level itself was compared within each sparsity mechanism, five of the six dataset-mechanism combinations showed significant differences. The exception was MovieLens recent-history retention, where performance did not change systematically across 50%, 25%, and 10% retention:

$$\chi^2(2) = 1.80, \qquad p = 0.407, \qquad W = 0.09. \tag{4}$$

Table IV provides a compact summary of key statistical results.

These statistical findings concern systematic differences across the evaluated aggregate model-condition results rather than significance at the level of individual users.

F. Cross-Dataset Robustness

Despite the large difference in dataset density, the full-data model rankings were highly similar. The Spearman correlation between MovieLens and Amazon baseline NDCG@10 rankings was

$$\rho = 0.939. \tag{5}$$

The agreement was much weaker when models were ranked according to their relative degradation under severe sparsity. At nominal 10% retention, the cross-dataset robustness correlations were $\rho = 0.467$ for random per-user retention, $\rho = -0.152$ for recent-history retention, and $\rho = 0.309$ for early-profile retention. None of these severe-sparsity robustness correlations was statistically significant after correction for multiple comparisons.

Supporting metrics produced similar model orderings. Across all conditions, Spearman correlations between NDCG@10 and Recall@10 rankings ranged from 0.927 to 0.988 on MovieLens and from 0.952 to 1.000 on Amazon. Agreement between

TABLE III
MODEL-WISE NDCG@10 AT NOMINAL 10% RETENTION. PARENTHESES SHOW RELATIVE CHANGE FROM THE FULL-DATA BASELINE (%); STOCHASTIC MODELS ARE THREE-SEED MEANS.

Panel A: MovieLens-1M

Model	Base	Random	Recent	Early
SASRec	.1191	.0265 (−77.7)	.0547 (−54.0)	.0213 (−82.1)
GRU4Rec	.1126	.0229 (−79.7)	.0515 (−54.3)	.0137 (−87.8)
BERT4Rec	.1067	.0174 (−83.7)	.0388 (−63.6)	.0129 (−87.9)
EASE	.0452	.0112 (−75.2)	.0461 (+2.0)	.0116 (−74.4)
MultiVAE	.0392	.0112 (−71.4)	.0415 (+5.9)	.0125 (−68.1)
LightGCN	.0381	.0112 (−70.6)	.0415 (+9.1)	.0095 (−74.9)
BPR	.0380	.0096 (−74.7)	.0368 (−3.3)	.0110 (−71.1)
ItemKNN	.0373	.0010 (−97.4)	.0295 (−21.0)	.0088 (−76.4)
NeuMF	.0320	.0121 (−62.2)	.0347 (+8.6)	.0110 (−65.5)
Pop	.0192	.0133 (−30.6)	.0172 (−10.2)	.0120 (−37.3)

Panel B: Amazon Video Games

Model	Base	Random	Recent	Early
SASRec	.0463	.0200 (−56.8)	.0278 (−40.0)	.0167 (−63.9)
GRU4Rec	.0449	.0206 (−54.2)	.0212 (−52.7)	.0165 (−63.2)
BERT4Rec	.0391	.0149 (−62.0)	.0145 (−63.1)	.0121 (−69.1)
EASE	.0406	.0041 (−89.8)	.0087 (−78.5)	.0039 (−90.4)
MultiVAE	.0346	.0120 (−65.2)	.0125 (−64.0)	.0111 (−67.9)
LightGCN	.0359	.0069 (−80.9)	.0080 (−77.8)	.0064 (−82.1)
BPR	.0355	.0044 (−87.5)	.0048 (−86.5)	.0042 (−88.3)
ItemKNN	.0241	.0022 (−90.7)	.0041 (−83.0)	.0020 (−91.8)
NeuMF	.0243	.0153 (−37.0)	.0158 (−35.2)	.0131 (−46.1)
Pop	.0163	.0157 (−3.2)	.0163 (+0.05)	.0148 (−9.1)

NDCG@10 and MRR@10 ranged from 0.867 to 1.000 and from 0.927 to 1.000 respectively. The appendix values confirm that the principal ordering is not specific to NDCG@10.

V. DISCUSSION

The results show that sparse-feedback performance cannot be explained by interaction quantity alone. Although recommendation quality generally decreased as less training feedback was retained, the extent of this degradation depended strongly on which interactions remained, which model was used, and the characteristics of the dataset. This distinction is important because two conditions with the same nominal retention level could produce substantially different outcomes.

A. Effect of Decreasing User Feedback

Regarding the effect of decreasing user feedback, the experiments show that reducing available user feedback generally decreases recommendation performance, but the relationship is not uniform. Random per-user and early-profile retention produced severe degradation at nominal 10% on both datasets.

Recent-history retention behaved differently. On MovieLens, no statistically significant systematic change was detected across retention levels ($p = 0.407$), and several non-sequential models matched or slightly exceeded their full-history NDCG@10 when only recent interactions were retained. This does not establish equivalence or show that removing data improves recommendation; the observed pattern is consistent with recent interactions being more predictive of the next held-out interaction in this dataset.

Amazon did not show the same tolerance to severe history reduction. At nominal 10%, approximately 9.76 interactions per user remained on MovieLens but only 1.18 on Amazon. The shorter retained histories are consistent with this greater sensitivity, while the accompanying loss of item exposure provides another contributor.

These findings are consistent with previous work showing that data sparsity can substantially affect recommender performance [2], while extending the comparison beyond overall sparsity level by explicitly controlling the temporal location of retained feedback.

B. Importance of Temporal Position

With respect to temporal position, retained feedback clearly matters. Recent-history retention significantly outperformed both random per-user and early-profile retention at every evaluated level on both datasets, with a particularly strong distinction on MovieLens under severe sparsity.

This finding is broadly consistent with Betello et al. [5], who showed that sequential recommender performance is sensitive to where interactions are removed from temporally ordered histories. Their experiments found particularly strong effects when interactions close to the end of a sequence were perturbed. Although the experimental protocols differ, both studies indicate that interactions should not be treated as equally informative simply because they belong to the same user history.

The poor performance of early-profile retention should also be interpreted carefully. It is a cold-start-like condition because only a small early portion of a known user's history is available, but it is not a true new-user cold-start experiment. In addition, the retained interactions are temporally distant from the test target. Its degradation may therefore reflect both reduced feedback quantity and stale preference information. The present experiments demonstrate that early interactions are generally less effective for the chosen next-interaction task, but they do not isolate the causal contribution of preference drift.

Sparsification also affected item exposure. As interactions were removed, some held-out test items became globally absent from the retained training data, particularly on Amazon. At early-profile 10%, this occurred for approximately 13.2% of Amazon test targets compared with 2.3% on MovieLens. The measured degradation therefore reflects both reduced user-profile information and, in some conditions, reduced opportunity for models to learn about particular items.

C. Model Performance and Robustness

In comparing model performance and robustness, SASRec led 17 of the 20 dataset-condition combinations and GRU4Rec led the remaining three. Friedman tests confirm a systematic model effect, but post-hoc analysis does not support treating SASRec as superior to every alternative; its aggregate rank did not differ significantly from GRU4Rec on either dataset.

Their absolute strength did not make them immune to the substantial sparse-feedback losses reported in Table III. "Recent is better" therefore compares retention mechanisms; it does not imply that severe sparsity has little effect on every model.

Greater architectural complexity did not automatically produce greater robustness. EASE was competitive at the Amazon baseline and under MovieLens recent-history sparsity, whereas NeuMF and LightGCN did not consistently outperform simpler

TABLE IV
SUMMARY OF KEY STATISTICAL RESULTS.

Analysis	MovieLens-1M	Amazon Video Games
Overall model effect	$\chi^2(9) = 63.03,\ p < .001,\ W = .700$	$\chi^2(9) = 59.22,\ p < .001,\ W = .658$
Mechanism effect, 50%	$p < .001,\ W = .73$	$p < .001,\ W = .84$
Mechanism effect, 25%	$p < .001,\ W = .91$	$p < .001,\ W = .73$
Mechanism effect, 10%	$p < .001,\ W = .76$	$p < .001,\ W = .91$
Retention effect, Recent	$p = .407,\ W = .09$	$p < .001,\ W = .84$

alternatives. This supports the need for tuned baselines and controlled evaluation [3], [4], [13]. Absolute quality must also be separated from relative robustness: Amazon Pop barely changed under Recent 10%, but SASRec still achieved much higher NDCG@10. A weakly personalised model may simply have less performance to lose, so stability alone does not imply better recommendations.

D. Cross-Dataset Agreement

Within the evaluated models, full-data rankings transferred strongly between datasets, but nominal 10% relative-robustness rankings did not, with none of the three correlations statistically significant after correction. Baseline accuracy and robustness are therefore related but distinct. The history-length contrast is one plausible explanation, but additional datasets are needed to test its generality.

E. Limitations and Future Work

The following considerations define the scope of the controlled evaluation and identify natural extensions of the study. The study applies controlled sparsification after 5-core filtering and therefore does not represent the full natural cold-start population. Early-profile retention is also cold-start-like rather than true new-user cold start, since users remain known and the retained interactions become increasingly distant from the test target. Random retention used a single fixed mask, ensuring that the same retained interactions were used consistently across model comparisons, while the minimum-one rule led realised retention to exceed the nominal level, particularly on Amazon. Because sparsification also reduced item exposure in some conditions, the measured degradation captures the combined practical effect of less user-profile information and less evidence about some target items, rather than a purely user-side effect. The use of multiple training seeds provides a view of run-to-run variability for stochastic models, so small score differences are interpreted cautiously. The aggregate statistical analysis is designed to assess systematic differences across the evaluated model-condition results, consistent with the stated level of inference.

A natural extension would be to repeat the random sparsification with multiple independent masks, allowing the sensitivity of the observed robustness patterns to the particular retained interactions to be quantified. Retaining per-user recommendation ranks would additionally enable paired analyses at the user level, complementing the aggregate model-condition comparisons used here. Further experiments could examine true new-user and new-item cold-start settings, additional recommendation domains, and datasets with different history-length distributions. It would also be useful to compare the frozen baseline-selected configurations used here with models retuned specifically for sparse conditions, providing a clearer view of how much performance can be recovered through sparsity-aware optimisation.

VI. CONCLUSION

This study examined how recommender systems respond when the amount and temporal position of available user feedback are systematically reduced. Ten recommender systems were evaluated on MovieLens-1M and Amazon Video Games under full-data conditions and random per-user, recent-history, and early-profile retention. Recommendation quality generally declined as feedback became sparse, but the extent of degradation depended strongly on which interactions remained. Recent-history retention usually preserved substantially more utility than equally sized random or early-profile subsets, with MovieLens showing greater tolerance to recent-history reduction than Amazon.

SASRec and GRU4Rec achieved the strongest absolute performance across most conditions, but greater architectural complexity did not consistently imply greater robustness. Sparse-feedback behaviour also differed between datasets, showing that strong full-data performance does not necessarily translate to similar robustness under limited feedback. Overall, the study demonstrates that recommender systems should be evaluated not only by how much interaction data remains, but also by which parts of the user history remain available.

The principal contribution is a controlled cross-paradigm evaluation showing that sparse-feedback robustness depends jointly on interaction quantity, temporal position, model choice, and dataset characteristics. Future work should extend the evaluation to additional domains using independent sparsity masks, sparse-condition retuning, and user-level paired comparisons.

ACKNOWLEDGEMENTS

I would like to thank my supervisor, Dr Hairong Bau, for her guidance, support, and valuable feedback throughout this research project. I also thank the supervision group for their helpful discussions and feedback, and the Mathematical Sciences Support team for providing the mscluster high-performance computing resources used in this work.

REFERENCES

[1] Y. Hu, Y. Koren, and C. Volinsky, "Collaborative filtering for implicit feedback datasets," in *Proceedings of the IEEE International Conference on Data Mining*, 2008, pp. 263–272.

[2] J. F. G. da Silva, N. N. de Moura Junior, and L. P. Caloba, "Effects of data sparsity on recommender systems based on collaborative filtering," in *Proceedings of the International Joint Conference on Neural Networks*, 2018, pp. 1–8.

[3] M. Ferrari Dacrema, P. Cremonesi, and D. Jannach, "Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches," in *Proceedings of the ACM Conference on Recommender Systems*, 2019, pp. 101–109.

[4] Z. Sun, D. Yu, H. Fang, J. Yang, X. Qu, J. Zhang, and C. Geng, "Are We Evaluating Rigorously? Benchmarking Recommendation for Reproducible Evaluation and Fair Comparison," in *Proceedings of the ACM Conference on Recommender Systems*, 2020, pp. 23–32.

[5] F. Betello, F. Siciliano, P. Mishra, and F. Silvestri, "Investigating the robustness of sequential recommender systems against training data perturbations," in *Advances in Information Retrieval: Proceedings of ECIR*, ser. Lecture Notes in Computer Science, vol. 14609, 2024, pp. 205–220.

[6] D. Jannach and L. Chen, "Improving methodological standards in recommender systems offline evaluation," *ACM Transactions on Recommender Systems*, vol. 4, no. 3, pp. 1–15, 2026, art. no. 34e.

[7] S. Rendle, C. Freudenthaler, Z. Gantner, and L. Schmidt-Thieme, "BPR: Bayesian personalized ranking from implicit feedback," in *Proceedings of the Conference on Uncertainty in Artificial Intelligence*, 2009, pp. 452–461.

[8] M. Quadrana, P. Cremonesi, and D. Jannach, "Sequence-aware recommender systems," *ACM Computing Surveys*, vol. 51, no. 4, Art. no. 66, 2018.

[9] B. Hidasi, A. Karatzoglou, L. Baltrunas, and D. Tikk, "Session-based recommendations with recurrent neural networks," in *Proceedings of the International Conference on Learning Representations*, 2016.

[10] W.-C. Kang and J. McAuley, "Self-attentive sequential recommendation," in *Proceedings of the IEEE International Conference on Data Mining*, 2018, pp. 197–206.

[11] F. Sun, J. Liu, J. Wu, C. Pei, X. Lin, W. Ou, and P. Jiang, "BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer," in *Proceedings of the ACM International Conference on Information and Knowledge Management*, 2019, pp. 1441–1450.

[12] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and powering graph convolution network for recommendation," in *Proceedings of the International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020, pp. 639–648.

[13] Z. Sun, H. Fang, J. Yang, X. Qu, H. Liu, D. Yu, Y.-S. Ong, and J. Zhang, "DaisyRec 2.0: Benchmarking recommendation for rigorous evaluation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 7, pp. 8206–8226, 2023.

[14] W. Krichene and S. Rendle, "On sampled metrics for item recommendation," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2020, pp. 1748–1757.

[15] D. Gusak, A. Volodkevich, A. Klenitskiy, A. Vasilev, and E. Frolov, "Time to split: Exploring data splitting strategies for offline evaluation of sequential recommenders," in *Proceedings of the ACM Conference on Recommender Systems*, 2025, pp. 874–883.

[16] F. Betello, A. Purificato, F. Siciliano, G. Trappolini, A. Bacciu, N. Tonellotto, and F. Silvestri, "A reproducible analysis of sequential recommender systems," *IEEE Access*, vol. 13, pp. 5762–5772, 2025.

[17] F. M. Harper and J. A. Konstan, "The MovieLens datasets: History and context," *ACM Transactions on Interactive Intelligent Systems*, vol. 5, no. 4, Art. no. 19, 2015.

[18] Y. Hou, J. Li, X. Fu, Z. He, A. Yan, X. Chen, and J. McAuley, "Bridging language and items for retrieval and recommendation: Benchmarking LLMs as semantic encoders," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2026, pp. 3251–3265.

[19] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, "Item-based collaborative filtering recommendation algorithms," in *Proceedings of the International Conference on World Wide Web*, 2001, pp. 285–295.

[20] H. Steck, "Embarrassingly shallow autoencoders for sparse data," in *Proceedings of the World Wide Web Conference*, 2019, pp. 3251–3257.

[21] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua, "Neural collaborative filtering," in *Proceedings of the International Conference on World Wide Web*, 2017, pp. 173–182.

[22] D. Liang, R. G. Krishnan, M. D. Hoffman, and T. Jebara, "Variational autoencoders for collaborative filtering," in *Proceedings of the World Wide Web Conference*, 2018, pp. 689–698.

[23] W. X. Zhao *et al.*, "RecBole: Towards a unified, comprehensive and efficient framework for recommendation algorithms," in *Proceedings of the ACM International Conference on Information and Knowledge Management*, 2021, pp. 4653–4664.

[24] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of IR techniques," *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002.

[25] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.

[26] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

APPENDIX A
SELECTED HYPERPARAMETERS

Table V lists the material selected parameters rather than unchanged RecBole defaults. Each model was tuned separately by dataset on the full-data baseline using validation NDCG@10; the listed configuration was then frozen across all sparse conditions. Trainable models used a learning rate of 10^{-3}, at most 100 epochs, and early stopping after 10 validation checks without improvement unless the model did not use iterative training.

TABLE V
SELECTED MODEL CONFIGURATIONS BY DATASET. ONLY PARAMETERS MATERIAL TO MODEL CAPACITY OR REGULARISATION ARE SHOWN.

Model	MovieLens-1M	Amazon Video Games
Pop	No learned model hyperparameters	No learned model hyperparameters
ItemKNN	$k = 50$, shrinkage $= 0$	$k = 200$, shrinkage $= 0$
BPR	embedding $= 64$, weight decay $= 10^{-6}$	embedding $= 64$, weight decay $= 10^{-6}$
EASE	regularisation $= 500$	regularisation $= 100$
NeuMF	MF embedding $= 64$, MLP embedding $= 32$, layers $[64, 32, 16, 8]$, dropout $= 0$	MF embedding $= 64$, MLP embedding $= 64$, layers $[64, 32, 16, 8]$, dropout $= 0$
MultiVAE	latent $= 200$, hidden $[600]$, dropout $= 0.5$	latent $= 200$, hidden $[600]$, dropout $= 0.5$
GRU4Rec	embedding $= 64$, hidden $= 128$, one layer, dropout $= 0.2$	embedding $= 64$, hidden $= 64$, one layer, dropout $= 0.2$
SASRec	hidden/embedding $= 64$, two layers, two heads, hidden dropout $= 0.1$, attention dropout $= 0.2$	hidden/embedding $= 64$, one layer, two heads, hidden dropout $= 0.2$, attention dropout $= 0.2$
BERT4Rec	hidden $= 64$, two layers, two heads, hidden dropout $= 0.1$, attention dropout $= 0.2$, mask ratio $= 0.2$	hidden $= 64$, two layers, two heads, hidden dropout $= 0.2$, attention dropout $= 0.2$, mask ratio $= 0.2$
LightGCN	embedding $= 64$, one layer, regularisation $= 10^{-5}$	embedding $= 64$, two layers, regularisation $= 10^{-5}$

APPENDIX B
SUPPORTING METRICS

Tables VI and VII report Recall@10 and MRR@10 for the full-data baseline and the three nominal 10% conditions. Values are means across seeds for stochastic models and single-run results for deterministic models. HitRate@10 is omitted because it is identical to Recall@10 under the single-held-out-target evaluation.

TABLE VI
MOVIELENS-1M SUPPORTING METRICS AT BASELINE AND NOMINAL 10% RETENTION.

	Recall@10				MRR@10			
Model	Baseline	Random	Recent	Early	Baseline	Random	Recent	Early
SASRec	0.2239	0.0635	0.1258	0.0454	0.0873	0.0154	0.0332	0.0141
GRU4Rec	0.2100	0.0478	0.1028	0.0294	0.0830	0.0155	0.0360	0.0091
BERT4Rec	0.2125	0.0363	0.0776	0.0290	0.0748	0.0117	0.0272	0.0082
EASE	0.0903	0.0250	0.0897	0.0244	0.0317	0.0071	0.0329	0.0077
MultiVAE	0.0812	0.0248	0.0849	0.0258	0.0267	0.0072	0.0285	0.0086
LightGCN	0.0796	0.0235	0.0845	0.0200	0.0257	0.0076	0.0287	0.0064
BPR	0.0792	0.0221	0.0736	0.0234	0.0258	0.0059	0.0257	0.0073
ItemKNN	0.0756	0.0023	0.0549	0.0169	0.0259	0.0005	0.0217	0.0063
NeuMF	0.0671	0.0262	0.0695	0.0231	0.0214	0.0079	0.0242	0.0074
Pop	0.0394	0.0305	0.0356	0.0242	0.0131	0.0083	0.0116	0.0083

TABLE VII
AMAZON VIDEO GAMES SUPPORTING METRICS AT BASELINE AND NOMINAL 10% RETENTION.

	Recall@10				MRR@10			
Model	Baseline	Random	Recent	Early	Baseline	Random	Recent	Early
SASRec	0.0963	0.0403	0.0604	0.0327	0.0311	0.0139	0.0178	0.0119
GRU4Rec	0.0850	0.0395	0.0413	0.0323	0.0328	0.0149	0.0152	0.0118
EASE	0.0727	0.0081	0.0162	0.0071	0.0308	0.0030	0.0064	0.0029
BERT4Rec	0.0743	0.0282	0.0282	0.0244	0.0285	0.0108	0.0103	0.0083
LightGCN	0.0659	0.0131	0.0149	0.0121	0.0268	0.0050	0.0059	0.0047
BPR	0.0650	0.0088	0.0095	0.0081	0.0265	0.0031	0.0034	0.0030
MultiVAE	0.0644	0.0246	0.0251	0.0232	0.0256	0.0083	0.0087	0.0075
NeuMF	0.0462	0.0301	0.0302	0.0274	0.0177	0.0109	0.0115	0.0088
ItemKNN	0.0439	0.0045	0.0082	0.0041	0.0180	0.0015	0.0028	0.0013
Pop	0.0320	0.0305	0.0321	0.0278	0.0116	0.0113	0.0116	0.0109

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☐
2. Sourcing Related Work (summarising, identifying sources) ☑
3. Methods and Experiment Design (experiment setup, model tuning) ☑
4. Data Analysis (presentation, coding, interpretation) ☑
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☑
7. Presentation (rendering graphics, formatting) ☑
8. Editing (grammar, readability) ☑
9. Writing (text generation, document structuring) ☐
10. Citation Formatting (structuring, organising) ☑

If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**
ChatGPT 5.6	Understanding concepts; research and experiment planning; related-work support; checking analysis and interpretation; editing, formatting, and citation organisation.
OpenAI Codex	Code/script generation and refinement; assisting with data analysis; checking implementation correctness

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2652159

Candidate signature: 

Date: 27/10/2026